

Sca

■

$$M)^2$$

الهد

=

400000

لدراسا

21

T

400000

=

80,000

Suppo
se

a

5

$$\sigma = \sqrt{200000}$$

اور مانندن

group of 12

Ines been fanted, as

=

282.84

to Sales price

follows:-

Способев

5, 10

10111 1

13, 15 35 50,

55, 721

192 2047

8

3

215. Partition 25 them into three bins by

Each of the following methods. (a)

canal frequency Partitioning (b)

label - width Positioning (<)

Clusterin

g

Total me

me

215

of roads

12

-9

being =3

@equal

frequency

Partitioning

729

Chewads pabin = Total ·
rewards
entnere Rewards

TO

3

If bonus

5

all

(6)

Eormal with
Partitioning

1

Bin

Bin 1

Values

5, 10, 11, 12

15, 35, 50, 55

Bin

2

Bim

3

72

(

9212041215

(b) . Equal widths Partitioning

width = Ironimum

No.

어

Minimum

bins

Bin

1

الملح

Internal:-

Bin 175

دوست تلا

تل المسى

$$1=5 - 7$$

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$$2 = 75-44$$

$$3 \Rightarrow 145 - 215$$

Interna

1

5

74

أكرم

L

Dest

Values

5, 10, 11, 13,
15, 35, 50, 55,

72

2

75-144

92

215

10153

(2)

Clustering

145-215 204

Cluster (Bin)

Cluske 1

Cluster

ن مسلسلہ

5,10, 11, 13, 15

2

35, 50, 5517 2 7 9 2

Cluster 3

204,215.

tested

Q. Suppose a hospitalized
leaned the are and
body fat
data

for

18 Randomly
selected

?

adults with the
following Exult

(a) Calmbate

the

mean

median and

Standard deviation of age and 1.

for

1

(b) Duane the

вопрок bom
plats for

cost and

%. fat.

(c) Namdize the two
Variable bared.

я

on

2-

бюле поетові зава

(d)

Draw a

Suether
phot

and a q-q plot

on there had Variable.

bared

(c) Coltulate the casilation
couffriant.

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negativally

Co-related

Оле

Are

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=2

23 + 23 + 27 + 27 + 39,

41 + 7 + 9 + 50

+

52 + 54 + 54

+ 56 + 57 + 58 + 58 +

60 + 61

816

18

18

= 46 · 43

1. fat =

518

=

18

28-78

Median

ade

50+52

=51

2

7334

العايسلمك

إلى

30-2 +31 :21=30.70

1. fat = 30:2+31 ; 2

2

5.1 to Jandard

demnation



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$$N = 1 \ll X$$

M

$$= (23 - 9.5)^2 + (2)$$

0.70
%

аде ∅ 12-85 . / .

fat_=) 8.99

Age

Min = 23, Q1

=

Median = 51, 39, Median

$Q1 = 57$

$\text{Max} = 61$

x

Age

Bit

4071-

Mindyiz

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24

3a

24

(J)

Q1:39

Med

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Mare

45

50.

55/8065

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35°

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29 47 49 41.47 49 50 52 54 56 57
58 60

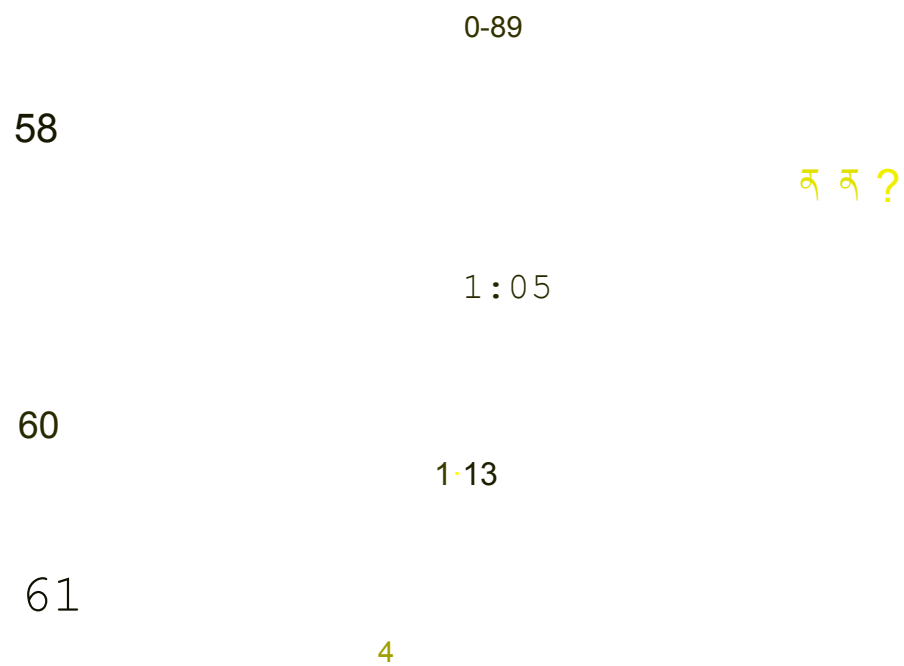
(c) Scatter
plot

egy

Stamplat

Satan phot





Copulation haefficient

P

Exy.

m2

R =

3

jn & xz ({ x) 2] [^

$$\{y^2=\{y\}]\dot{z}$$

$$Ex\; Ey$$

$$Ex=836$$

$$n = \text{Age} \sim \sim \sim$$

$$9=18$$

$$Ey--514.1$$

270

$$G = \; V \; fat. \; Ex = 41798$$

$$Eyz \; 16,368$$

الف

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2280153

$$3506X1461.70$$

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Similarity and diimidarit

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disimilarity

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$$d(i-j) = \underline{P-m}$$

2)

Cantigamy table

1

S
9V

Jum.

12+5

P

of Sum

978

$$\begin{array}{c}
 t \\
 + t \\
 r+t \\
 P
 \end{array}$$

For Symmetric binary
Variable,

$$\begin{array}{c}
 1-t \\
 \text{Jamila d} (i, j) = \\
 x+5
 \end{array}$$

WBA

Asymmetric
binary
Variables,

7+8

matriz

q+r+s+t

d(iis
)

vαζ

$$9+8+8$$

...Janand
Coefficient

welov

Steamed

($\dot{i}, \dot{j}$
)

a

$$a+r+5$$

4)

zz

Some

انت

$$= 2$$

$$\begin{matrix} x- & M \\ x \end{matrix}$$

Find
ing

the distance measure
Min koneski distance

$h = 1,$

Mummation
distance

$$d(i, j) =$$
$$[$$

$h=2$

$d(i,j)$

+

+

21

the name is faid tobe
Coulledd

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-d (i , j)

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Activity-5

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(3, 5)

(1, 2)

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$$5 - 2$$

(x3xg)

z

$$= 3,$$

31

(2322)

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3. (as
of)

(2.05
(112)

d=611

(210)

(3.5

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(112)

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x

d=6

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5.317

(4, 5) ,
(3, 5)

1 + -5 1 ~ + / 3 - 5 /
2 (1 7)

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JANV

(1) (x=22) 27 (3, 5)

(112)

57

=>

max

3-11 15-

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 $(x3x3) = (2, 0)$ $(1, 2)$ $\Rightarrow$ mar

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(P)^T
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Cosine Similarity

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$$\cos (dz.de) =$$
$$(dgods) / |||$$

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1

the indicates vector dot
Product

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length of

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(iv)

$\sim 32 + (\sim 10)$

$(3, 5)$

max

J

all

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(IN) - TRAX

(415) (IL)

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x

max 3, 3

(~) 422 (41)

(3.5)

C.4

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(5, 0, 3, 0, 2, 0, 0, 2,
0, 0)

dž

d2 =

(31912 10, 1, 1,
0,1,0,

3+0

d=d_y = 5+ 3 + 0 +

$$0 + 3 * 2 + 0 *$$

19

$$+ 2$$

$$0 * 0 + 0$$

라

$$1 \text{ to } 1 \text{ to of } 2 * 1 +$$

$$L 1$$

$$+ 6 + 0$$

15

$$+ \text{년}$$

$$+ 2 +$$

$$25/1$$

$$\parallel \text{ dell } - 5 \text{ to}$$

$$+ g''$$

$$+ 0 + 2 \text{ to to}$$

сол

02723

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+

22 то 2 того

22+02

= 25+ @ + 9 + 4 + 4

b

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42

||

4j

~

6.48

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-8/1

to 9 to + 4 to +1 +1 to +1+

-√17

4·12/1

0+ /